All code is in `Scripts/`, one script per question. The general methods are in the root module `High_Dimensional_Regression.py`. Each script saves its figures in `Figures/` and its numbers in `Results/`. I used Python, with scikit-learn supplying the Lasso solver.

Q1. Learning the Gaussian graphical model

I generated $n = 300$ independent observations from $N_{400}(0, \Theta^{-1})$, using the concentration matrix exactly as given: diagonal 1, first off-diagonals 0.5, second off-diagonals 0.25, and zero elsewhere. The zero mean is an assumption. The smallest eigenvalue of Θ is 0.25005, so it is positive definite. I used a Cholesky factor $LL^T = \Theta$ and generated $x = L^{-T}z$, where $z \sim N_{400}(0, I)$.

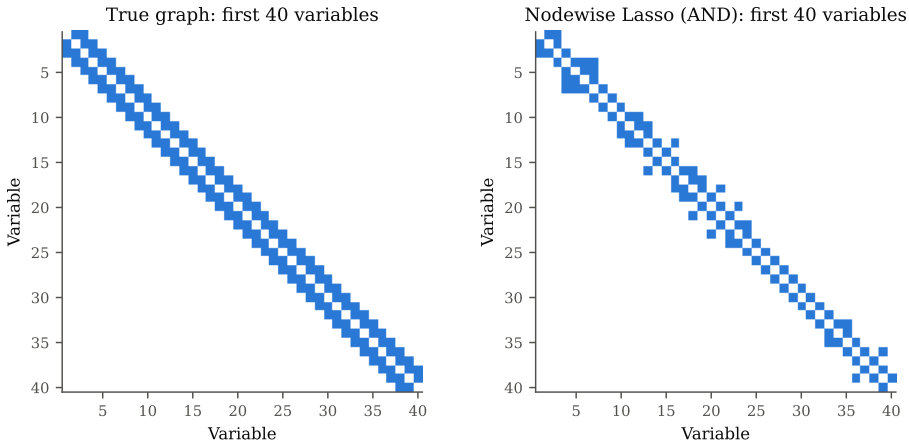
I used nodewise Lasso (Meinshausen and Bühlmann, 2006). For a Gaussian distribution, an edge joins variables j and k exactly when $\theta_{jk} \neq 0$. After centring and scaling each column to mean square 1, I regressed each standardized column Z_j on all the others:

$$\hat{\gamma}_j = \arg\min_{\gamma} \left\{ \frac{\|Z_j - Z_{-j}\gamma\|^2}{2n} + \lambda_G \|\gamma\|_1 \right\}, \qquad \lambda_G = \sqrt{\frac{2\log p}{n}} = 0.19986.$$

This penalty was fixed without using the true graph. I included an edge when both regressions selected the pair (the AND rule), treating coefficients below 10^{-8} in magnitude as zero. The OR rule includes an edge if either regression selects it. Each undirected edge is counted once.

Setting / result	**AND**	**OR**
Observations / variables / seed	300 / 400 / 546	
True edges	797	
Estimated edges	551	650
True positives / false positives / missed edges	492 / 59 / 305	551 / 99 / 246
Precision / recall	0.893 / 0.617	0.848 / 0.691

The AND graph recovers much of the banded pattern, but misses 305 true edges. The OR rule recovers more edges and also adds more false ones. Precision is the fraction of selected edges that are true; recall is the fraction of the 797 true edges recovered. The figure shows the first 40 variables; the table uses all 400.



Q2. High-dimensional regression inference

Data and fitting

I used the same predictor matrix as in Q1 and generated

$$y_i = 1 + 2x_{i1} + 2.5x_{i2} + 3x_{i3} + 3.5x_{i4} + 4x_{i5} + \epsilon_i.$$

The remaining 395 slopes are zero. The assignment does not specify the errors; I used independent $\epsilon_i \sim N(0,1)$, independent of the predictors, with seed 547.

I used desparsified Lasso (van de Geer et al., 2014). Let $Z_{ij} = (x_{ij} - \bar{x}_j)/s_j$, with $s_j^2 = n^{-1}\sum_i (x_{ij} - \bar{x}_j)^2$, and let $y_c = y - \bar{y}$. I first fitted

$$\hat{a} = \arg\min_a \left\{ \frac{\|y_c - Za\|^2}{2n} + \lambda_Y \|a\|_1 \right\}.$$

Five-fold cross-validation chose λ_Y by minimum prediction error over 40 logarithmically spaced values from 0.01 to 0.5. Each fold fitted its own centring and scaling on the training observations. The intercept was not penalized.

Bias correction and uncertainty

I used a separate nodewise penalty λ_N for inference. It was selected by five-fold cross-validation, averaging prediction error over all 400 nodewise regressions, from 20 logarithmically spaced values from 0.01 to 0.3. This choice uses only the predictors. Graph selection in Q1 and the approximate inverse needed here have different purposes, so their penalties need not agree.

For each nodewise fit, let $r_j = Z_j - Z_{-j}\hat{\gamma}_j$ and define

$$\hat{\tau}_j^2 = \frac{r_j^T r_j}{n} + \lambda_N \|\hat{\gamma}_j\|_1, \qquad M_{jj} = \frac{1}{\hat{\tau}_j^2}, \quad M_{j,-j} = -\frac{\hat{\gamma}_j^T}{\hat{\tau}_j^2}, \quad Q = ZM^T.$$

I kept M as this rowwise approximate inverse of the sample correlation matrix. With $e = y_c - Z\hat{a}$, the corrected coefficients and their standard errors in the original units are

$$\hat{b}_j = \frac{\hat{a}_j + Q_j^T e/n}{s_j}, \qquad \widehat{\mathrm{se}}_j = \frac{\hat{\sigma}\|Q_j\|}{n s_j}, \qquad \hat{\sigma}^2 = \frac{e^T e}{n - \hat{s} - 1},$$

where $\hat{s}$ is the number of nonzero slopes in the initial Lasso. The residual degrees-of-freedom correction estimates the noise variance; it is not an exact unbiased estimate after selection.

I formed pointwise 95% intervals $\hat{b}_j \pm 1.96\widehat{\mathrm{se}}_j$ and two-sided normal p-values for $H_{0j} : \beta_j = 0$. For testing all 400 slopes, I used $p_{j,\mathrm{adj}} = \min(1, 400p_j)$ and rejected at 0.05. The intervals are marginal, not simultaneous. This inference relies on a high-dimensional normal approximation and sufficiently small remaining regularization bias; it is not an exact finite-sample guarantee.

Results

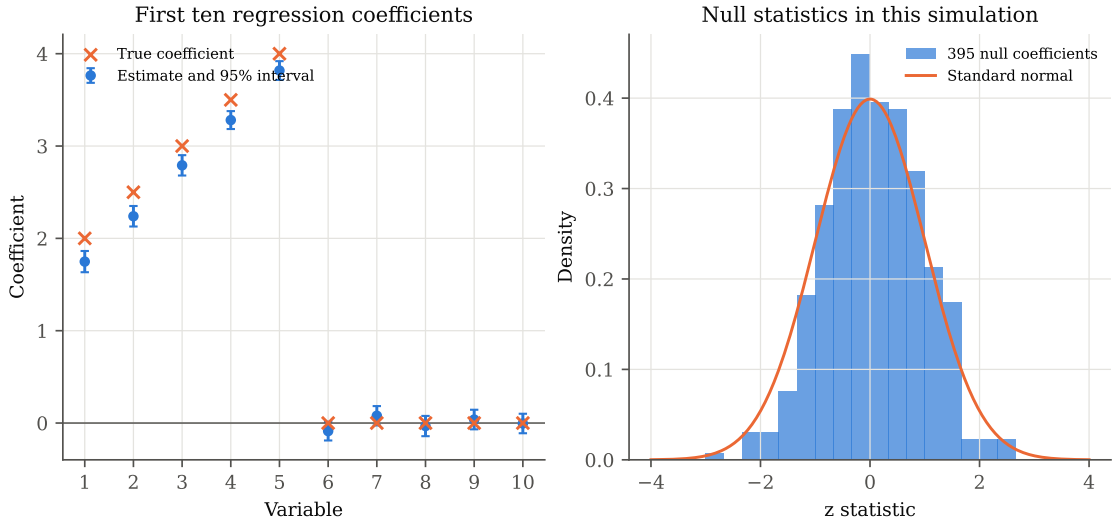
Setting / result	Value
Observations / variables / design seed	300 / 400 / 546
Cross-validation folds / shuffle seed	5 / 546
Selected λ_Y / selected λ_N	0.08219 / 0.08569
Initial nonzero slopes / estimated error SD	50 / 1.078
Solver tolerance / maximum iterations	10^{-8} / 10,000
Bonferroni-significant variables	**1, 2, 3, 4, 5**

The five nonzero coefficients are shown below. Very small p-values are given as bounds; a numerical underflow to zero does not mean that the mathematical p-value is zero.

Variable	True	Estimate	SE	95% interval	p_{adj}
1	2.0	1.748	0.059	[1.633, 1.863]	$< 10^{-190}$
2	2.5	2.240	0.057	[2.129, 2.351]	$< 10^{-190}$
3	3.0	2.791	0.056	[2.681, 2.901]	$< 10^{-190}$
4	3.5	3.281	0.050	[3.184, 3.378]	$< 10^{-190}$
5	4.0	3.818	0.052	[3.716, 3.919]	$< 10^{-190}$

The adjusted tests identify all five nonzero slopes and no null slopes. Without adjustment, 12 of the 395 null slopes are significant at 0.05. The intercept estimate $\hat{b}_0 = \bar{y} - \bar{x}^T\hat{b}$ is 0.970, compared with the generating value 1.

Interval check: 383 of the 395 null intervals contain zero, but none of the five nonzero coefficients lies in its interval. Their estimates are 0.182–0.260 below the true values. The bias correction leaves substantial shrinkage in this correlated design, so the nominal 95% intervals are not reliable for these five coefficients in this run. The overall count, 383 out of 400, would hide that failure. These counts describe one simulated dataset, not repeated-simulation coverage.



Saved results

The saved `q2.json` contains estimates, standard errors, intervals and raw and adjusted p-values for every slope. The simulated predictors and concentration matrix are in `q1_data.npz`; the response, true slopes and errors are in `q2_data.npz`.

References

STAT 546, Fall 2026. Homework 5, Questions 1–2. The concentration matrix, regression coefficients and requested methods are taken from the assignment.

N. Meinshausen and P. Bühlmann (2006). High-dimensional graphs and variable selection with the Lasso. *The Annals of Statistics*, 34(3), 1436–1462. DOI: 10.1214/009053606000000281.

S. van de Geer, P. Bühlmann, Y. Ritov and R. Dezeure (2014). On asymptotically optimal confidence regions and tests for high-dimensional models. *The Annals of Statistics*, 42(3), 1166–1202. DOI: 10.1214/14-AOS1221.